

ZARIF FIDA CHOWDHURY

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SUMMARY

Computer Science senior graduating August 2026 focused on secure AI systems, application security, cloud/security engineering, and SOC-ready fundamentals. Built Capstone GPT, a full-stack AI assistant with explicit backend security controls: server-side LLM API key handling, session authentication, bcrypt password hashing, 256-bit tokens, and server-side input validation. Hands-on foundation across software security, network security, data security, Linux, SIEM research, AWS, Docker, and secure SDLC.

SKILLS

Security/AppSec: OWASP Top 10, threat modeling, SAST/DAST, penetration testing, secure SDLC, input validation, authentication/session security, applied cryptography

Security Tools: Wireshark, tcpdump, Nmap, Metasploit, Burp Suite, OpenSSL, Kali Linux

Cloud/Systems: AWS, Docker, Linux/Bash, Git, GitLab, REST APIs, MySQL, PostgreSQL

AI/Software: RAG pipelines, Dify, LLM API integration, Python, Java, C++, C, PHP, JavaScript, SQL

PROJECTS

Capstone GPT - Secure AI Systems Project, Miami University

Spring 2026

- Built a full-stack RAG AI assistant for Senior Design students using PHP 8, MySQL, JavaScript/Tailwind, and Dify; delivered solo and managed development through GitLab issues, branches, and merge requests.
- Kept the LLM API key server-side behind PHP middleware, creating a single auditable control point for AI calls, validation, prompt composition, and activity logging.
- Implemented session-based authentication with bcrypt password hashing and 256-bit tokens; validated input server-side before requests reached the Dify workflow.
- Built PHP 8 REST middleware with 8 endpoints over a normalized 5-table MySQL schema, supporting chat logs, feedback, instructor analytics aggregation, and CSV export.
- Designed a Dify knowledge-retrieval workflow grounded in course materials, with topic-aware routing and structured student-profile context for more relevant responses.

EXPERIENCE

CSE Student Ambassador - Miami University

Oct 2025 - Present

Oxford, OH

- Represent the Computer Science & Software Engineering department at recruitment events, campus visits, and outreach activities.
- Engage with technical and non-technical audiences to explain program details, student experience, and computing pathways.

Cybersecurity Intern - SMART Technologies Ltd.

Jun 2024 - Aug 2024

Dhaka, Bangladesh

- Completed Cisco Linux Essentials, building hands-on Linux administration and command-line proficiency.
- Researched and delivered an internal presentation on Security Information and Event Management (SIEM) systems.
- Trained on enterprise networking practices, including port forwarding and network configuration.

Software Developer Intern - SMART Technologies Ltd.

Jun 2023 - Aug 2023

Dhaka, Bangladesh

- Developed backend prototypes in Python for data processing pipelines.
- Designed and optimized PostgreSQL schemas for production workloads.
- Presented technical solutions to stakeholders, translating requirements into deliverables.

EDUCATION

Miami University - B.S. Computer Science

Expected Aug 2026

Oxford, OH

Security Coursework: CYB 331 Software Security - secure SDLC, threat modeling, OWASP Top 10, CSRF/XSS/injection, input validation, SAST/DAST, fuzz testing, code reviews, software supply chain security.

CYB 334 Network Security - TLS/HTTPS/SSH, Kerberos, X.509/PKI, 802.1X, DNSSEC, DoH, SPF/DKIM/DMARC, MITM, ARP attacks, IDS/IPS, VPN, firewalls, wireless LAN security.

CYB 236 Data Security - AES, RSA, ElGamal, digital signatures, PKI, Diffie-Hellman, TLS, HMAC/CMAC, authenticated encryption, hash functions, cryptanalysis, side-channel and fault-injection attacks.

Additional Coursework: Computer Networks, Database Systems, Systems I & II, Object-Oriented Programming, Data Abstractions & Structures, Algorithms, Machine Learning.

ACTIVITIES

Cybersecurity Club, Miami University | Sep 2022 - Present